Caden Phillips

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MPhys Astrophysics student at the University of Liverpool, specialising in compact objects, time-domain astrophysics, and computational modelling. Research experience in Fisher matrix investigation, TOV solver building and analysis, hybrid equation of state construction, rapidly rotating neutron star simulations (RNS and LORENE), and X-ray time-domain analysis of NICER data. Comfortable with building and working with pipelines from numerical solutions to data processing, with strong Python and scientific computing skills.

Education

MPhys Astrophysics, University of Liverpool *Expected 2027*

Projected First Class · Year 3: 73.5%

Academic Advisor: Dr. Christopher Copperwheat

Selected modules: Computational Modelling, Relativity and Cosmology, Advanced Observational Astrophysics, Physics of Galaxies, Physics of Planets

Full transcript available upon request.

Research Experience

Internship — Institute of Nuclear Physics (IFJ PAN), Kraków, Poland *July–August 2025*

Supervisor: Dr. David Álvarez Castillo

Project: Physical Inputs for Simulations of Gravitational Wave Emissions from Compact Stars within the Einstein Telescope Detection Range

- Applied a Fisher Information Matrix framework to estimate tidal deformability uncertainties and assess their propagation in binary neutron star systems.
- Developed and implemented a Tolman–Oppenheimer–Volkoff (TOV) solver to compute mass–radius and tidal deformability relations for hybrid equations of state.
- Processed a dataset of 62 hybrid equations of state, generating M–R and Λ–M curves ($\sim$400 stellar configurations per EoS) and analysing stable and unstable branches.
- Compared model predictions with GW170817 50% and 90% credible regions to assess equation-of-state viability.
- Presented project results jointly with a fellow intern at the IFJ PAN Summer Internship Symposium 2025.

Field Work — Teide Observatory, Tenerife, Spain *June 2025*

Supervisor: Dr. Daniel Harman

- Conducted observations using the 0.8 m IAC-80 telescope, as well as a Celestron C14 Edge HD system with a QHY-163M/C CMOS detector.
- Worked in a team to identify and observe targets, such as M107, M13, Abell 39.
- Calibrated and reduced multi-band CCD data to construct composite images and Hertzsprung–Russell diagrams.

Independent Study — HEASARC/NICER Analysis Pipeline *Winter 2025–Spring 2026*
Self-directed observational analysis project

- Developed a Python-based pipeline integrating HEASoft tools via Bash to automatically download and organise archival NICER X-ray observations from the HEASARC database.
- Automated event filtering and calibration prior to light-curve extraction using scripted XSELECT workflows across multiple ObsIDs.
- Produced calibrated X-ray light curves and diagnostic plots to enable time-domain comparison of source variability across multiple observing epochs.

Independent Study — Rotating Neutron Star Modelling *Autumn–Winter 2025*
Self-directed follow-up research after IFJ PAN internship

- Investigated the effects of stellar spin on neutron star structure, extending previous TOV-based models which neglect rotation.
- Used RNS and LORENE frameworks to model rapidly rotating neutron stars under different equations of state.
- Developed a C++ program to automate RNS and LORENE runs for a single tabulated EoS, exploring how rotational frequency affects mass, radius, and relativistic stability limits.

Research Outputs

- **Presentation:** *Gravitational Wave Emissions from Compact Stars within the Einstein Telescope Detection Range.* IFJ PAN Summer Internship Symposium 2025. [slides]
- **Report:** *Physical Inputs for Simulations of Gravitational Wave Emissions from Compact Stars within the Einstein Telescope Detection Range.* Institute of Nuclear Physics (IFJ PAN), Kraków, 2025. [PDF].
- **Pipeline:** *Using HEASoft to Analyse Archived NICER Data of tMSPs.* Independent Study, 2026. [Code].

Technical Skills

- **Programming:** Python (NumPy, SciPy, Matplotlib, Astropy), C++, Bash
- **Tools & software:** RNS, LORENE, Jupyter, ROOT, Git, LaTeX, HEASoft
- **Scientific methods:** TOV and hybrid EoS modelling, Fisher matrix estimation, numerical ODE solvers, gravitational wave parameter inference
- **Languages:** English (native), Portuguese (conversational)

Awards & Memberships

- Awarded competitive summer internship placement at IFJ PAN (Poland) through the PHYS309 Internship Module.
- Associate Member, Institute of Physics (IOP) *since 2023*.

Interests

Astrophysics (compact objects, neutron stars, gravitational waves); astrophotography; hiking.